

Experiment Number: **04**

Implementing a Four-input Priority Encoder

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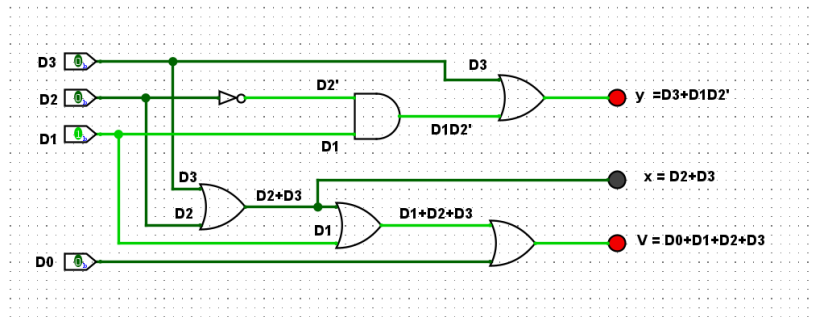


Figure 1: Four-input Priority Encoder using Logic Gates

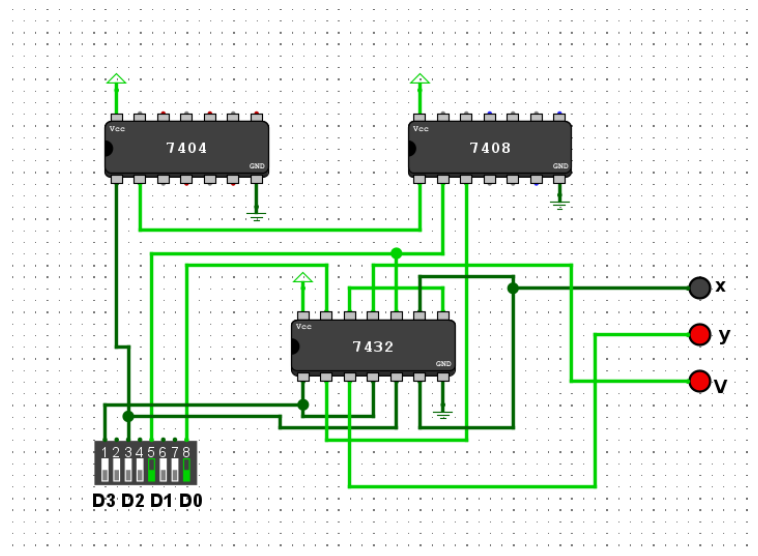


Figure 2: Four-input Priority Encoder using Logic IC's

```
1 module priority_encoder_4to2 (  
2     input [3:0] D,  
3     output reg [1:0] Y,  
4     output V  
5 );  
6  
7  
8 assign V = |D; // capital V matches output  
9  
10 always @(*) begin  
11     if(D[3])  
12         Y = 2'b11;  
13     else if (D[2])  
14         Y = 2'b10;  
15     else if (D[1])  
16         Y = 2'b01;  
17     else if (D[0])  
18         Y = 2'b00;  
19     else  
20         Y = 2'b00;  
21 end  
22  
23  
24 endmodule
```

```
---- 4 to 2 Priority Encoder ----  
D   | Y V  
0000 | 00 0  
0001 | 00 1  
0010 | 01 1  
0100 | 10 1  
1000 | 11 1  
1111 | 11 1  
tb.v:54: $finish called at 300000 (1ps)
```

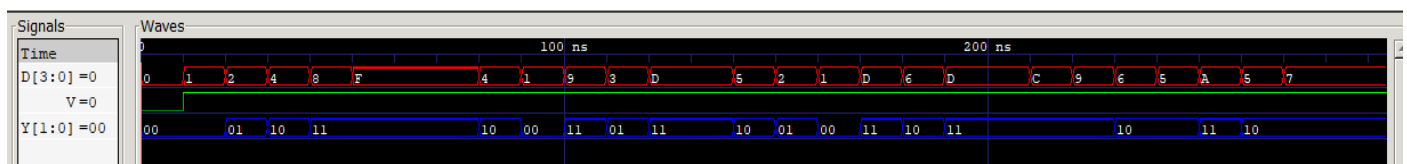


Figure 3: Verilog Code with corresponding Truth table and Signal Response for Four-input Priority Encoder